Facundo Mémoli

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Academic Appointments

Professor. Rutgers University. 2025–present.
Professor. The Ohio State University (OSU). 2020–2024.
Associate Professor. The Ohio State University. 2016–2020.
Assistant Professor. The Ohio State University. 2013–2016.
Assistant Professor. The University of Adelaide. 2011–2013.
Postdoctoral fellow. Stanford University, 2005–2011.

Education

University of Minnesota (UMN), Minneapolis, Minnesota USA. Ph.D., 2005.
Universidad de la República, Montevideo, Uruguay. M. Sci, 2001.
Universidad de la República, Montevideo, Uruguay. Electrical Engineering degree, 2001.

Grants

- NSF # DMS 2520735. Conference: Algebraic Topology: Methods, Computation, and Science 2025 (ATMCS11). $24,999. 2025–2026. Co-PI.
- NSF # DMS 2301359. Multiparameter Topological Data Analysis. $199,989. 2023–2027 (NCE).[1] PI.
- NSF # CCF 2310412. Graph Analysis: Integrating Metric and Topological Perspectives. $299,985. 2023–2027 (NCE). PI.
- NSF # IIS-1901360. Through synapses to spatial learning: a topological approach. $435,000. 2019–2026 (NCE). PI.
- United States–Israel Binational Science Foundation (BSF) #2020124. The filling radius and Topological Data Analysis. $140,000. 2021–2025. PI.
- NSF # CCF-1740761. TRIPODS: TGDA@OSU: Structure, Shape, and Dynamics in Data. $1.5M. 2017–2023 (NCE). PI.
- NSF # DMS-1547357. RTG: Algebraic Topology and Its Applications. $1.72M. 2016–2023 (NCE). Co-PI.
- NSF # CCF-1839356. TRIPODS+X:EDU: An MBI TGDA+Neuro for Undergraduates. Mathematical Biosciences Institute (MBI), OSU. $199,983. 2018–2021 (NCE). Co-PI.
- NSF # DMS-1723003. Topology of Functional Data on Random Metric Spaces, Graphs &Graphons. $180K. 2017–2020. PI.
- NSF # CCF-1839358. TRIPODS+X:RES: Thermodynamic Phases and Config. Space Topology. $300K. 2018–2022. Co-PI.

[1] No cost extension.

- NSF # DMS-1743943. CBMS Conference: Elastic Functional and Shape Data Analysis (EFSDA). $35,748. 2018. Co-PI.
- NSF # DMS-1613094. Conference on Topology, Geometry, and Data Analysis at OSU. $40,000. 2016–2017. Co-PI.
- NSF # CCF-1526513. Analyzing Complex Data with a Topological Lens. $399,999. 2015–2018. Co-PI.
- NSF # IIS-1422400. Robustness of spatial learning in flickering networks: the hippocampus. $230,176. 2014–2017. PI.
- Office of Naval Research (ONR) # N00014-09-1-0783. Metric Geometry and Category Theory for Data Fusion. $666,813. 2009–2011. Senior Personnel.
- ONR # N00014-10-1-0808. SIAM-IS 2010 workshop "Metric and Riemannian methods in Shape Analysis". $29,291. Co-PI.

Visiting positions, fellowships, and prizes

- Korea Advanced Institute of Science & Technology (KAIST); Visiting Professor, Spring 2026.
- Australian National University; Visiting Professor, Spring 2026.
- Sydney Mathematical Research Institute; Visiting Professor, Spring 2026.
- Paul R. Halmos – Lester R. Ford Mathematical Association of America (MAA) award for Expository Mathematical Writing, 2025.
- Institute for Computational and Experimental Research in Mathematics (ICERM), Brown University; Visiting Professor, Summer 2017.
- Institute for Mathematics and its Applications (IMA), UMN; Visiting Professor Fall 2013. Semester program: Scientific and Engineering applications of Algebraic Topology.

Papers in Journals

68. S. Chen, S. Lim, F. Mémoli, Z. Wan, Y. Wang. Weisfeiler-Lehman Meets Optimal Transport: Theory, Algorithms, and Implications. *Journal of Machine Learning Research* (2026; to appear).
67. C. Eastwood, A. M. Ellison, M. Gómez, F. Mémoli. The topology of curvature sets of S^1. *Topology and its Applications* 391, 109921 (2026).
66. H. Adams, J. Bush, N. Clause, F. Frick, M. Gómez, M. Harrison, R. A. Jeffs, E. Lagoda, S. Lim, F. Mémoli, M. Moy, N. Sadovek, M. Superdock, D. Vargas, Q. Wang, L. Zhou. Gromov-Hausdorff distances, Borsuk-Ulam theorems, and Vietoris-Rips complexes. *Algebraic & Geometric Topology* (2026; to appear).
65. S. Arya, A. Auddy, R. A. Clark, S. Lim, F. Mémoli, D. Packer. The Gromov-Wasserstein distance between spheres. *Foundations of Computational Mathematics* 26, 75–130 (2026).
64. M. Bauer, F. Mémoli, T. Needham, M. Nishino. The Z-Gromov-Wasserstein distance. *Journal of Machine Learning Research* 26(291), 1–57 (2025). Selected for poster presentation at NeurIPS 2026. https://neurips.cc/virtual/2026/loc/atlanta/poster/156658.
63. F. Mémoli, B. Vose, R. C. Williamson. Geometry and Stability of Supervised Learning Problems. *Journal of Machine Learning Research* 26(221), 1–99 (2025).
62. A. Balitskiy, B. Coskunuzer, F. Mémoli. Geometric Bounds for Persistence. *Transactions of the American Mathematical Society* 378(12), 8437–8486 (2025).
61. N. Clause, T. K. Dey, F. Mémoli, B. Wang. Meta-diagrams for 2-parameter persistence. *Discrete & Computational Geometry* 74, 872–898 (2025).
60. F. Mémoli, L. Zhou. Persistent homotopy groups of metric spaces. *Journal of Topology and Analysis* 17(5), 1481–1542 (2025).

59. F. Mémoli, L. Zhou. Ephemeral persistence features and the stability of filtered chain complexes. *Journal of Computational Geometry* 15(2), 258–328 (2024).
58. F. Mémoli, T. Needham. Comparison results for Gromov-Wasserstein and Gromov-Monge distances. *ESAIM: Control, Optimisation and Calculus of Variations* 30, 78 (2024).
57. S. Lim, F. Mémoli. Classical multidimensional scaling on metric measure spaces. *Information and Inference: A Journal of the IMA* 13(2), iaae007 (2024).
56. M. Gómez, F. Mémoli. Curvature sets over persistence diagrams. *Discrete & Computational Geometry* 72(1), 91–180 (2024).
55. T. Dey, W. Kim, F. Mémoli. Computing generalized rank invariant for 2-parameter persistence modules via zigzag persistence and its applications. *Discrete & Computational Geometry* 71(1), 67–94 (2024).
54. F. Mémoli, A. Stefanou, L. Zhou. Persistent cup product structures and related invariants. *Journal of Applied and Computational Topology* 8(1), 93–148 (2024).
53. W. Kim, F. Mémoli. Extracting persistent clusters in dynamic data via Möbius inversion. *Discrete & Computational Geometry* 71(4), 1276–1342 (2024).
52. W. Kim, F. Mémoli, A. Stefanou. Interleaving by parts: Join decompositions of interleavings and join-assemblage of geodesics. *Order* 41(2), 497–537 (2024).
51. M. Gómez, F. Mémoli. The Four Point Condition: An Elementary Tropicalization of Ptolemy's Inequality. *The American Mathematical Monthly* 131(3), 187–203 (2024).
50. H. Adams, F. Mémoli, M. Moy, Q. Wang. The persistent topology of optimal transport based metric thickenings. *Algebraic & Geometric Topology* 24(1), 393–447 (2024).
49. S. Lim, F. Mémoli, O. B. Okutan. Vietoris-Rips persistent homology, injective metric spaces, and the filling radius. *Algebraic & Geometric Topology* 24(2), 1019–1100 (2024).
48. W. Kim, F. Mémoli. Persistence over posets. *Notices of the American Mathematical Society* 70(8), 1214–1225 (2023).
47. F. Mémoli, A. Munk, Z. Wan, C. Weitkamp. The ultrametric Gromov-Wasserstein distance. *Discrete & Computational Geometry* 70(4), 1378–1450 (2023).
46. H. Lyu, F. Mémoli, D. Sivakoff. Sampling random graph homomorphisms and applications to network data analysis. *Journal of Machine Learning Research* 24(9), 1–79 (2023).
45. F. Mémoli, Z. Smith, Z. Wan. The Gromov-Hausdorff distance between ultrametric spaces: its structure and computation. *Journal of Computational Geometry* 14, 78–143 (2023).
44. S. Lim, F. Mémoli, Z. Smith. The Gromov-Hausdorff distance between spheres. *Geometry & Topology* 27(9), 3733–3800 (2023).
43. F. Mémoli, Z. Wan. Characterization of Gromov-type geodesics. *Differential Geometry and its Applications* 88, 102006 (2023).
42. S. Chowdhury, F. Mémoli. Distances and isomorphism between networks: stability and convergence of network invariants. *Journal of Applied and Computational Topology* 7(2), 243–361 (2023).
41. F. Mémoli, T. Needham. Distance distributions and inverse problems for metric measure spaces. *Studies in Applied Mathematics* 149(4), 943–1001 (2022).
40. F. Mémoli, Z. Wan, Y. Wang. Persistent Laplacians: Properties, algorithms and implications. *SIAM Journal on Mathematics of Data Science* 4(2), 858–884 (2022).
39. F. Mémoli, Z. Wan. On p-metric spaces and the p-Gromov-Hausdorff Distance. *p-Adic Numbers, Ultrametric Analysis and Applications* 14(3), 173–223 (2022).
38. C. Cai, W. Kim, F. Mémoli, Y. Wang. Elder-rule-staircodes for augmented metric spaces. *SIAM Journal on Applied Algebra and Geometry* 5(3), 417–454 (2021).
37. W. Kim, F. Mémoli. Generalized persistence diagrams for persistence modules over posets. *Journal of Applied and Computational Topology* 5(4), 533–581 (2021).
36. G. Carlsson, F. Mémoli, S. Segarra. Robust hierarchical clustering for directed networks: an axiomatic approach. *SIAM Journal on Applied Algebra and Geometry* 5(4), 675–700 (2021).

35. U. Bauer, C. Landi, F. Mémoli. The Reeb graph edit distance is universal. *Foundations of Computational Mathematics* 21, 1441–1464 (2021).
34. W. Kim, F. Mémoli. Spatiotemporal persistent homology for dynamic metric spaces. *Discrete & Computational Geometry* 66, 831–875 (2021).
33. F. Mémoli, O. B. Okutan. Quantitative simplification of filtered simplicial complexes. *Discrete & Computational Geometry* 65, 554–583 (2021).
32. S. Chowdhury, N. Clause, F. Mémoli, J. Sánchez, Z. Wellner. New families of stable simplicial filtration functors. *Topology and its Applications* 279, 107254 (2020).
31. F. Mémoli, O. B. Okutan. Reeb posets and tree approximations. *Discrete Mathematics* 343(2), 111658 (2020).
30. D. H. Martínez, F. Mémoli, W. Mio. The shape of data and probability measures. *Applied and Computational Harmonic Analysis* 48(1), 149–181 (2020).
29. S. Chowdhury, F. Mémoli. The Gromov-Wasserstein distance between networks and stable network invariants. *Information and Inference: A Journal of the IMA* 8(4), 757–787 (2019).
28. H. Hang, F. Mémoli, W. Mio. A topological study of functional data and Fréchet functions of metric measure spaces. *Journal of Applied and Computational Topology* 3(4), 359–380 (2019).
27. F. Mémoli, K. Singhal. A primer on persistent homology of finite metric spaces. *Bulletin of Mathematical Biology* 81, 2074–2116 (2019).
26. A. Elchesen, F. Mémoli. The reflection distance between zigzag persistence modules. *Journal of Applied and Computational Topology* 3, 185–219 (2019).
25. P. Frosini, F. Mémoli, C. Landi. The Persistent Homotopy Type Distance. *Homotopy, Homology, and Applications* 21(2), 231–259 (2019).
24. S. Chowdhury, F. Mémoli. A functorial Dowker theorem and persistent homology of asymmetric networks. *Journal of Applied and Computational Topology* 2, 115–175 (2018).
23. S. Chowdhury, B. Dai, F. Mémoli. The importance of forgetting: Limiting memory improves recovery of topological characteristics from neural data. *PLOS ONE* 13(9), e0202561 (2018).
22. S. Chowdhury, F. Mémoli. Explicit geodesics in Gromov-Hausdorff space. *Electronic Research Announcements* 25, 48–59 (2018).
21. F. Mémoli, A. Sidiropoulos, V. Sridhar. Quasimetric embeddings and their applications. *Algorithmica* 80, 3803–3824 (2018).
20. G. Carlsson, F. Mémoli, A. Ribeiro, S. Segarra. Hierarchical clustering of asymmetric networks. *Advances in Data Analysis and Classification* 12, 65–105 (2018).
19. G. Carlsson, F. Mémoli, A. Ribeiro, S. Segarra. Admissible hierarchical clustering methods and algorithms for asymmetric networks. *IEEE Transactions on Signal and Information Processing over Networks* 3(4), 711–727 (2017).
18. L. Cosmo, E. Rodola, A. Albarelli, F. Mémoli, D. Cremers. Consistent partial matching of shape collections via sparse modeling. *Computer Graphics Forum* 36(1), 209–221 (2017).
17. A. Babichev, D. Ji, F. Mémoli, Y. A. Dabaghian. A topological model of the hippocampal cell assembly network. *Frontiers in Computational Neuroscience* 10, 50 (2016).
16. G. Carlsson, F. Mémoli. Classifying clustering schemes. *Foundations of Computational Mathematics* 13, 221–252 (2013).
15. Y. Dabaghian, F. Mémoli, L. Frank, G. Carlsson. A Topological Paradigm for Hippocampal Spatial Map Formation Using Persistent Homology. *PLOS Computational Biology* 8(8), e1002581 (2012).
14. F. Mémoli. Some properties of Gromov-Hausdorff distances. *Discrete & Computational Geometry* 48, 416–440 (2012).
13. T. Le, F. Mémoli. Local scales on curves and surfaces. *Applied and Computational Harmonic Analysis* 33(3), 401–437 (2012).
12. F. Mémoli. Gromov-Wasserstein distances and the metric approach to object matching. *Foundations of Computational Mathematics* 11, 417–487 (2011).

11. F. Mémoli. A spectral notion of Gromov-Wasserstein distance and related methods. *Applied and Computational Harmonic Analysis* 30(3), 363–401 (2011).
10. M. Ovsjanikov, Q. Mérigot, F. Mémoli, L. Guibas. One point isometric matching with the heat kernel. *Computer Graphics Forum* 29(5), 1555–1564 (2010).
9. G. E. Carlsson, F. Mémoli. Characterization, stability and convergence of hierarchical clustering methods. *Journal of Machine Learning Research* 11(Apr), 1425–1470 (2010).
8. F. Chazal, D. Cohen-Steiner, L. J. Guibas, F. Mémoli, S. Y. Oudot. Gromov-Hausdorff stable signatures for shapes using persistence. *Computer Graphics Forum* 28(5), 1393–1403 (2009).
7. G. Singh, F. Mémoli, T. Ishkhanov, G. Sapiro, G. Carlsson, D. L. Ringach. Topological analysis of population activity in visual cortex. *Journal of Vision* 8(8), article 11, 1–18 (2008).
6. C. Moenning, F. Mémoli, G. Sapiro, N. Dyn, N. A. Dodgson. Meshless geometric subdivision. *Graphical Models* 69(3–4), 160–179 (2007).
5. F. Mémoli, G. Sapiro. A theoretical and computational framework for isometry invariant recognition of point cloud data. *Foundations of Computational Mathematics* 5, 313–347 (2005).
4. F. Mémoli, G. Sapiro. Distance Functions and Geodesics on Submanifolds of $\mathbb{R}^d$ and Point Clouds. *SIAM Journal on Applied Mathematics* 65(4), 1227–1260 (2005).
3. F. Mémoli, G. Sapiro, P. Thompson. Implicit brain imaging. *NeuroImage* 23, S179–S188 (2004).
2. F. Mémoli, G. Sapiro, S. Osher. Solving variational problems and partial differential equations mapping into general target manifolds. *Journal of Computational Physics* 195(1), 263–292 (2004).
1. F. Mémoli, G. Sapiro. Fast computation of weighted distance functions and geodesics on implicit hypersurfaces. *Journal of Computational Physics* 173(2), 730–764 (2001).

Papers in Peer Reviewed Conferences and Workshops

34. *Ephemeral persistence features and the stability of filtered chain complexes*. With L. Zhou. SoCG 2023.
33. *A generalization of the persistent Laplacian to simplicial maps*. With A. Gülen, Z. Wan and Y. Wang. SoCG 2023.
32. *Meta-Diagrams for 2-Parameter Persistence*. With N. Clause, T. Dey and B. Wang. SoCG 2023.
31. *Weisfeiler-Lehman meets Gromov-Wasserstein*. With S. Chen, S. Lim, Z. Wan and Y. Wang. ICML 2022.
30. *Computing Generalized Rank Invariant for 2-Parameter Persistence Modules via Zigzag Persistence and its Applications*. With W. Kim and T. Dey. SoCG 2022.
29. *Persistent Cup-Length*. With M. Contessotto, S. Stefanou and L. Zhou. 2021. SoCG 2022.
28. *The Reeb Graph Edit Distance is Universal*. With U. Bauer and C. Landi. SoCG 2020.
27. *Elder-rule-staircodes for Augmented Metric Spaces*. With C. Cai, W. Kim, and Y. Wang. SoCG 2020.
26. *The Wasserstein Transform*. With Z. Wan and Z. Smith. ICML 2019: The 36th International Conference on Machine Learning.
25. *Formigrams: Clustering Summaries of Dynamic Data*. With W. Kim. CCCG: The Canadian Conference on Computational Geometry, 2018.
24. *Persistent Path Homology of Asymmetric Networks*. With S. Chowdhury. SODA 2018.
23. *Topological Analysis of Nerves, Reeb Spaces, Mappers, and Multiscale Mappers*. With T. Dey and Y. Wang. SOCG 2017.
22. *Improved error bounds for tree representation of metric spaces data*. With S. Chowdhury and Z. Smith. NIPS 2016.
21. *Quasimetric embeddings and applications*. With A. Sidiropoulos and V. Sridhar. ICALP 2016.
20. *Distances between directed networks and applications*. With S. Chowdhury. ICASSP, 2016.
19. *Multiscale Mapper: A Framework for Summarization of Data and Maps*. With T. Dey and Y. Wang. SODA 2016.

18. *Metric Structures on Networks and Applications*. With S. Chowdhury. Allerton, 2015.
17. *Hierarchical Quasi-Clustering Methods for Asymmetric Networks*. With G. Carlsson, A. Ribeiro and S. Segarra. ICML 2014.
16. *Alternative Axiomatic Constructions for Hierarchical Clustering of Asymmetric Networks*. With G. Carlsson, A. Ribeiro, and S. Segarra. In IEEE Global Conference on Signal and Information Processing, 2013.
15. *Multiscale covariance fields, local scales, and shape transforms*. With D. Díaz and W. Mio. In Geometric Science of Information, 2013.
14. *Hierarchical Clustering Methods and Algorithms for Asymmetric Networks*. In Asilomar Conference on signals, systems, and computers, 2013.
13. *Axiomatic construction of hierarchical clustering in asymmetric networks*. With G. Carlsson, A. Ribeiro, and S. Segarra. In Proc. Int. Conf. Acoustics Speech Signal Process, 2013.
12. *Some ideas for formalizing clustering methods*. With G. Carlsson. In NIPS 2009 workshop "Clustering: Science or Art? Towards Principled Approaches".
11. *Multiparameter Hierarchical Clustering methods*. With G. Carlsson. Proceedings of the 11th IFCS Biennial Conference and 33rd Annual Conference of the Gesellschaft für Klassifikation e.V., Dresden, March 13–18, 2009. Studies in Classification, Data Analysis, and Knowledge Organization Springer, Berlin-Heidelberg-New York.
10. *Spectral Gromov-Wasserstein distances for shape matching*. ICCV workshop on Non-Rigid Shape Analysis and Deformable Image Alignment (NORDIA'09), Kyoto, Japan. September 2009.
9. *Stability of Clustering methods*. With G. Carlsson. Mathematisches Forschungsinstitut Oberwolfach Report No. 29/2008, "Computational Algebraic Topology".
8. *Gromov-Hausdorff distances in Euclidean spaces*. IEEE-CVPR, workshop on Non-Rigid Shape Analysis and Deformable Image Alignment (NORDIA'08), Alaska, June 2008.
7. *Topological Methods for the Analysis of High Dimensional Data Sets and 3D Object Recognition*. With G. Carlsson and G. Singh. PBG 2007 (Point Based Graphics). Prague, Czech Republic, September 2007.
6. *On the use of Gromov-Hausdorff distances for Shape Comparison*. PBG 2007 (Point Based Graphics). Prague, Czech Republic, September 2007.
5. *Geometric Surface and Brain Warping via Geodesic Minimizing Lipschitz Extensions*. With P. Thompson and G. Sapiro. Proceedings of the First International Workshop on Mathematical Foundations of Computational Anatomy - Geometrical and Statistical Methods for Modelling Biological Shape Variability, MICCAI 2006. Copenhagen, Denmark, October 2006.
4. *Comparing Point Clouds*. With G. Sapiro. Symposium on Geometric Processing, 2004. Nice, France.
3. *Distance Functions and Geodesics on Point Clouds*. With G. Sapiro. In 2nd IEEE Workshop on Variational, Geometric and Level Set Methods in Computer Vision In Conjunction with the 9th IEEE International Conference in Computer Vision (ICCV'03), October 12, 2003, Nice, France.
2. *Solving Variational Problems and PDEs mapping into General Target Manifolds*. With G. Sapiro and S. Osher. In 2nd IEEE Workshop on Variational, Geometric and Level Set Methods in Computer Vision In Conjunction with the 9th IEEE International Conference in Computer Vision (ICCV'03), October 12, 2003, Nice, France.
1. *Difusión Anisotrópica*. With A. Pardo & A. Ribeiro. IV Simposio Iberoamericano de Reconocimiento de Patrones (SIARP'99), La Habana, Cuba, 1999.

Chapters in Books and overview papers

7. *Analysis of Dynamic Graphs and Dynamic Metric Spaces via Zigzag Persistence*. With W. Kim and Z. Smith. In proceedings of the 2018 Abel Symposium.
6. *Distances Between Datasets*. In "Modern Approaches to Discrete Curvature". Springer LNM. 2017.
5. *The Gromov-Wasserstein distance: a brief overview*. *Axioms* 3, 335–341 (2014).

4. *The Gromov-Hausdorff distance: a brief tutorial on some of its quantitative aspects*. *Actes des rencontres du CIRM* 3(1), 89–96 (2013). Courbure discrete : theorie et applications.

3. *Metric structures on datasets: stability and classification of algorithms*. *Lecture Notes in Computer Science* 6855, 1–33 (2011). Proceedings of the 14th international conference on Computer analysis of images and patterns. Invited paper, CAIP 2011.

2. *Computing with Point Cloud Data*. With G. Sapiro. In *Statistics and Analysis of Shapes (Modeling and Simulation in Science, Engineering and Technology)*. H. Krim and A. Yezzi, editors. Birkhäuser, May 2006.

1. *Variational Problems and PDEs on Implicit Surfaces*. With M. Bertalmío, L.-T. Cheng, G. Sapiro & S. Osher. In *Geometric Level Set Methods in Imaging, Vision and Graphics*. S. Osher and N. Paragios, editors. Springer-Verlag, August 2002.

Preprints

- *Critical Diameters and Vietoris-Rips Filtrations*. With Q. Wang. 2026. https://arxiv.org/abs/2609.33009.
- *Vietoris–Rips Coindex Thresholds for Round Spheres: Spherical Joins and Chromatic Obstructions for Gromov–Hausdorff Distances*. With S. Lim and Q. Wang. 2026. https://arxiv.org/abs/2609.07778.
- *The Gromov-Hausdorff Distance Between Consecutive Spheres*. With D. Kim and S. Lim. 2026. https://arxiv.org/abs/2608.13264.
- *Metric Geometry of Lebesgue, Wasserstein, and Gromov-Wasserstein Spaces: Submetries, Curvature, and Geodesics*. With M. Bauer, T. Needham, and M. Nishino. 2026. https://arxiv.org/abs/2608.11680.
- *Discrimination of Dynamic Data via Curvature Sets*. With N. Belova, M. Goldberg, S. Raghunath, and A. Xie. 2026. https://arxiv.org/abs/2603.04624.
- *Persistence and Topological Complexity*. With L. Zhou. 2025. https://arxiv.org/abs/2506.17888.
- *The G-Gromov-Hausdorff Distance and Equivariant Topology*. With S. Lim. 2025. https://arxiv.org/abs/2506.15414.
- *ℓ^p-Stability of Weighted Persistence Diagrams*. With A. B. Gülen and A. Patel. 2025. https://arxiv.org/abs/2504.11694.
- *Grassmannian Persistence Diagrams: Special Properties in the 1-Parameter Setting*. With A. B. Gülen and Z. Wan. 2025. https://arxiv.org/abs/2504.06077.
- *Embedding-Projection Correspondences for the estimation of the Gromov-Hausdorff distance*. With Z. T. Smith. 2024. https://arxiv.org/abs/2407.03295.
- *Computation of degree-1 persistent homology on larger point-clouds using the Reduced Vietoris-Rips filtration*. With M. Koyama, K. Turner and V. Robins. 2024. https://arxiv.org/abs/2307.16333.
- *The Generalized Rank Invariant: Möbius invertibility, Discriminating Power, and Connection to Other Invariants*. With N. Clause and W. Kim. 2024. https://arxiv.org/abs/2207.11591.
- *Grassmannian Persistence Diagrams*. With A. Gülen and Z. Wan. 2023. https://arxiv.org/abs/2311.06870.
- *Extremal spherical polytopes and Borsuk's conjecture*. With M. Katz and Q. Wang. 2023. https://arxiv.org/abs/2301.13076.
- *Some results about the Tight Span of spheres*. With S. Lim, Z. Wan, and Q. Wang. 2021. https://arxiv.org/abs/2112.12646.
- *The Gaussian Transform*. With K. Jin and Z. Wan. 2020. https://arxiv.org/abs/2006.11698.
- *On the notion of weak isometry for finite metric spaces*. With A. De Gregorio, U. Fugacci, and F. Vaccarino. 2020. https://arxiv.org/abs/2005.03109.

- *Metric Representations of Network Data*. With S. Segarra, G. Carlsson, and A. Ribeiro. 2019. https://arxiv.org/abs/1911.00164.
- *Motivic Clustering Schemes for Directed Graphs*. With G. Vituri. 2019. https://arxiv.org/pdf/2001.00278.pdf.
- *Metric graph approximation of geodesic spaces*. With O. Okutan. September 2018. https://arxiv.org/abs/1809.05566.
- *The Distortion of the Reeb Quotient Map on Riemannian Manifolds*. With O. Okutan. January 2018. https://arxiv.org/abs/1801.01562.
- *Sketching and Clustering Metric Measure Spaces*. With K. Singhal and A. Sidiropoulos. January 2018. https://arxiv.org/abs/1801.00551.
- *Stable Signatures for Dynamic Graphs and Metric Spaces via Zigzag Persistent Homology*. With W. Kim. December 2017. https://arxiv.org/abs/1712.04064v4.
- *A Distance Between Filtered Spaces Via Tripods*. April 2017. https://arxiv.org/abs/1704.03965.
- *Persistent Clustering and a theorem of J.Kleinberg*. With G. Carlsson. February 2008. http://arxiv.org/abs/0808.2241.
- *Local Scale Selection for Exploratory Visualization and Analysis of Massive Datasets*. With G. Carlsson, D. Müellner, and G. Singh.

Professional Service

- Associate Editor for Orbita Matematicae. A Math Journal of UMALCA (Union Matemática de América Latina y el Caribe) https://www.umalca.org/orbita-mathematicae/.
- Organizer of BACTA 2025 (Bridging Algebraic Combinatorics, Topology and Applications) at Rutgers University. https://facundo-memoli.org/BACTA-2025.html.
- PC member, SoCG 2023 (Symposium on Computational Geometry). https://cs.utdallas.edu/SOCG23/socg.html.
- Co-organizer, AMS Math Research Communities on “Data Science at the Crossroads of Analysis, Geometry, and Topology”. 2022. http://www.ams.org/programs/research-communities/mrc-22.
- Co-organizer, Algebraic Topology and Topological Data Analysis: A conference in honor of Gunnar Carlsson. IMA, Minneapolis, August 2022. https://cse.umn.edu/ima/events/algebraic-topology-and-topological-data-analysis-conference-honor-gunnar-carlsson.
- Co-organizer, conference “Bridging Applied and Quantitative Topology”. May 2022. https://sites.google.com/view/applied-quantitative-topology/.
- Co-organizer, The Vietoris-Rips seminar. Bi-weekly online seminar, 2021–2022. https://sites.google.com/view/aatrn-vr-seminar/home.
- Co-organizer, workshop on Algebraic Combinatorics and Category Theory in Topological Data Analysis. March 2022. https://sites.google.com/view/tda-alg-combi-cat/home.
- Co-organizer, AAAI workshop on Optimal Transport and Structured Data Modeling. February 2022. https://ot-sdm.github.io/.
- Co-organizer, workshop on Optimal Transport, Topological Data Analysis and Applications to Shape and Machine Learning. MBI-OSU, July 2020.
- ICIAM 2019. Organizer of minisymposium on Geometric and Topological Data Analysis (w/ Yasu Hiraoka and Washington Mio).
- Co-organizer, TGDA@OSU 2018 Summer School. MBI-OSU, May 14–18, 2018.
- Co-organizer, TGDA@OSU Workshop on Theory and Foundations of TGDA. MBI-OSU, May 21–25, 2018.

- Co-organizer, special session on “Topology and Geometry in Data Analysis”. American Mathematical Society (AMS) Spring Central Sectional Meeting, OSU, 2018.
- Co-organizer, 8-week Summer@ICERM 2017 program on “Topological Data Analysis”.
- Co-organizer, special session on “Geometry and Topology in Image and Shape Analysis”. AMS Sectional Meeting, North Carolina State University (NCSU), 2016.
- Co-organizer, minisymposium on “Topology and Geometry Across Scales”. SIAM Imaging Science, 2016.
- Co-organizer, TGDA@OSU 2016 conference.
- Co-organizer, Topology and Geometry in Data Analysis seminar. OSU, 2013–2024.
- Co-organizer, workshop “Applications to Image Processing and Shape Analysis” (with Irina Kogan). SIAM Conference on Applied Algebraic Geometry, 2013.
- PC member, SMI 2011.
- Co-organizer, workshop “Metric and Riemannian methods in shape analysis” (with Mario Micheli and Martin Reuter). SIAM Imaging Science conference, Chicago, April 12–14, 2010.
- Area chair for Asian Conference on Computer Vision (ACCV) 2010.
- PC member, workshop on Non-Rigid Shape Analysis and Deformable Image Alignment (NORDIA), 2009 and 2010 (ICCV and CVPR).

Other service

Grant review and panels for NSF, Israel Science Foundation (ISF), etc.

Reviewing

Foundations of Computational Mathematics.
Transactions of the American Mathematical Society.
Journal of Applied and Computational Topology.
Probability Theory and Related Fields.
Algebraic & Geometric Topology.
Journal of Mathematical Analysis and Applications.
Proceedings of the American Mathematical Society.
Annals of Statistics.
Transactions on Algorithms.
Advances in Mathematics.
Discrete & Computational Geometry.
Topology and its Applications.
Mathematics of Computation.
IEEE Signal Processing Letters.
SIAM Journal on Applied Algebra and Geometry.
Proceedings of the National Academy of Sciences.
Information and Inference.
IEEE Transactions on Image Processing.
Journal of Computational Physics.
Applied Mathematics Letters.
IEEE Transactions on Pattern Analysis and Machine Intelligence.
IEEE Letters.
Advances in Data Analysis and Classification.
Journal of Machine Learning Research.

Conferences: ICML, NeurIPS, SoCG, CVPR, ICCV, SIGGRAPH.

Outreach activities

- *The Gromov-Hausdorff distance between spheres*. Glimpse Talk, Special Pre-enrollment Program in Mathematics. Rutgers University. August 20, 2026.
- Co-organizer, graduate student conferences on Geometry and Topology meet Data Analysis and Machine Learning. 2019 and 2021. https://tgda.osu.edu/gtdaml2019/ and https://gtdaml.wixsite.com/2021.
- Minicourse on *Network Data Analysis*. Centro de Investigación y de Estudios Avanzados del Instituto Politécnico Nacional (CINVESTAV), Mexico, January 2017.
- OSU *Invitation to Mathematics* lecture series, February 2016.

- Minicourse on *Introduction Applied Algebraic Topology and Data Analysis*. Universidade Estadual Paulista (UNESP), Rio Claro, October 2015.
- Minicourse on *Introduction Applied Algebraic Topology and Data Analysis*. University of Science, VietNam National University, Ho Chi Minh City, August 2015.
- OSU *Invitation to Mathematics* lectures, April 9 and 16, 2014.
- Minicourse on *Introduction to Mathematical Analysis Methods in Image Processing*. University of Science, VietNam National University, Ho Chi Minh City, August 1–5, 2011.
- Minicourse on *Object matching under invariances: a metric approach*. Centre International de Mathématiques Pures et Appliquées (CIMPA) school: Applied Mathematics and Engineering. Uruguay, 2010.

Ph.D. and MMS advising

Past

- Brantley Vose, Ph.D. student, OSU, 2021–2024. Now at Upstart.
- Musashi Koyama, OSU and Australian National University, 2019–2025. Now a postdoctoral fellow at Max Planck Institute of Molecular Cell Biology and Genetics (MPI-CBG), Dresden.
- Aziz Gülen, Ph.D. student, OSU, 2020–2024. Was a postdoctoral fellow at Duke University. Now at Google Research.
- Nate Clause, Ph.D. student, OSU, 2020–2024. Now in industry.
- Ling Zhou, Ph.D. student, OSU, 2018–2023. Was a postdoctoral fellow at Duke. Now tenure-track Assistant Professor at the University of Oklahoma.
- Mario Gomez, Ph.D. student, OSU, 2019–2023. Now a postdoctoral fellow at Florida State University (FSU).
- Samir Chowdhury, Ph.D. student, OSU, 2014–2019. Was a postdoctoral fellow at Stanford University. Now Staff Computer Vision Engineer at Bear Flag Robotics (John Deere).
- Zhengchao Wan, Ph.D. student, OSU, 2017–2022. Was a postdoctoral fellow at the University of California, San Diego (UCSD). Now tenure-track Assistant Professor at the University of Missouri.
- Kun Jin, Ph.D. student, OSU, 2018–2021. Now with Samsung Research.
- Woojin Kim, Ph.D. student, OSU, 2015–2020. Was a postdoctoral fellow at Duke. Now in a tenure-track position at KAIST, South Korea.
- Sunhyuk Lim, Ph.D. student, OSU, 2014–2021. Was a postdoctoral fellow at the Max Planck Institute for Mathematics in the Sciences (MPI-MiS), Leipzig. Now Assistant Professor at Sungkyunkwan University.
- Osman Okutan, Ph.D. student, OSU, 2016–2019. Was a postdoctoral fellow at Florida State University and then MPI-MiS, Leipzig. Now in industry at Clarios.
- Kritika Singhal, Ph.D. student, OSU, 2017–2020. Was a Senior Associate at Axtria. Now at DePuy Synthes.
- Qingsong Wang, Ph.D. student, OSU, 2019–2022. Was a postdoctoral fellow at the University of Utah. Now at UCSD.
- Guilherme Vituri, Ph.D. student, UNESP-Rio Claro, Brazil, 2017–2020. Now in industry.
- Marco Contessotto, Ph.D. student, UNESP-Rio Claro, Brazil, 2017–2022.
- Alexander Elchesen, MMS student, OSU, 2015–2017. Ph.D., University of Florida. Was a postdoctoral fellow at Colorado State University (CSU). Now Software Engineer (AI/ML) at Google.
- Xiao Zha, MMS student, OSU, 2017–2019.
- Ying Yin, MMS student, OSU, 2017–2019. Now at Wolfram.

Current

- Trung Nghia Nguyen, Ph.D. student. Rutgers, 2025–present.

Postdoctoral mentoring

- Arun Suresh, postdoctoral fellow, Rutgers, 2026–present.
- David Hien, postdoctoral fellow, Rutgers, 2025–present.
- Yiqing Cai, postdoctoral fellow, IMA, 2013. Now Data Scientist at Gro Intelligence.
- Tom Needham, postdoctoral fellow, OSU, 2016–2019. Now Associate Professor at Florida State University.
- Anastasios Stefanou, postdoctoral fellow, OSU, 2018–2021. Now in a tenure-track position at the University of Bremen, Germany.
- Ranthony Edmonds, postdoctoral fellow, OSU, 2020–2023. Was at Duke. Now Assistant Professor at OSU.
- Alexander McCleary, postdoctoral fellow, OSU, 2021–2023. Now at Montana State University.

Teaching

- Rutgers University. Gromov's Filling Radius and Persistent Homology. Fall 2026.
- Rutgers University. Differential geometry. Fall 2025.
- Rutgers University. Mathematics of Data Science. Spring 2025.
- The Ohio State University. Differential Geometry of Curves and Surfaces, 2024.
- The Ohio State University. Topics on Optimal Transport in Data Science, 2023.
- The Ohio State University. Riemannian Geometry, 2015, 2016, 2018, 2022.
- The Ohio State University. Topics on Mathematical Phylogenetics, 2020.
- The Ohio State University. Topics on Random Metric Spaces, 2019.
- The Ohio State University. Topics and Optimal Transport in Machine Learning and Shape Analysis. 2018.
- The Ohio State University. Topics on Network Data Analysis, 2016.
- The Ohio State University. Scientific Computing, 2015.
- The Ohio State University. Ordinary and Partial Differential Equations, 2014.
- The Ohio State University. Metric Geometry and Shape Matching, 2014.
- University of Adelaide, Australia. Introduction to Geometric Algorithms, 2011. Introduction to Machine Learning, 2012. Scientific computing, 2013. Puzzle Based Learning, 2013.
- Universidad de la República, Uruguay. Object matching under invariances: a metric approach (graduate course), 2010.
- Stanford University. Real Analysis, 2007. Shape Matching: A Metric Geometry Approach (graduate course), 2008.
- Universidad de la República, Uruguay. Sampling Theory and Signal Processing, Analog and Digital Communication Systems, Image Processing, 1999–2002.

Invited Talks and Courses

Keynote (∗) Plenary (∗∗) Course (♯)

Round Spheres as Benchmarks for Metric Matching: Current Progress and Open Problems. FoCM 2026, Computational Geometry and Topology workshop. University of Vienna, Vienna, Austria. July 10, 2026.

Curvature Sets over Persistence Diagrams. Order and Disorder in Random Geometries: Hyperuniformity and Topological Data Analysis. Schloss Reisensburg, Günzburg, Germany. July 7, 2026.

(∗) *Grassmannian Persistence Diagrams*. TORI Workshop. Wimereux, France. July 2, 2026.

(∗) *Persistent Laplacians and spectral refinements of persistent homology*. Discrete Laplacians 2026. MPI-CBG, Dresden, Germany. June 22, 2026.

Grassmannian Persistence Diagrams. Topology, Geometry, and Data Analysis Seminar. KAIST, Daejeon, Republic of Korea. May 15, 2026.

The G-Gromov-Hausdorff distance and Equivariant Topology. Topology, Geometry, and Data Analysis Seminar. KAIST, Daejeon, Republic of Korea. May 8, 2026.

The Gromov-Hausdorff distance between spheres. Pure & Applied Mathematics Seminar. University of Newcastle, Australia. March 13, 2026.

Distance distributions and inverse problems for metric measure spaces. Mathematical Physics Seminar. Rutgers University. February 12, 2026.

(∗) *La G-distancia de Gromov-Hausdorff y topología equivariante*. Jornadas de Topología de datos. Universidad de Sevilla. February 2026.

(∗) *Grassmannian Persistence diagrams*. ATMCS 2026. MPI-MiS, Leipzig. January 2026.

(∗) *The G-Gromov-Hausdorff distance and Equivariant Topology*. Special session at JMM 2026. Washington DC. January 2026.

(∗) *The Gromov-Hausdorff distance between spheres*. Coloquio Uruguayo de Matemáticas. Montevideo, Uruguay. December 2025.

(∗) *The Gromov-Hausdorff distance between spheres*. TDA conference at University of Missouri. November 2025.

(∗) *The G-Gromov-Hausdorff distance and Equivariant Topology*. TDA conference at University of Missouri. November 2025.

(∗∗) *The Gromov-Hausdorff distance between spheres*. 32nd Annual Fall Workshop on Computational Geometry. Queens College. November 2025.

The Gromov-Hausdorff distance between spheres. AMS Sectional meeting. Tulane University. October 2025.

The G-Gromov-Hausdorff distance and Equivariant Topology. AMS Sectional meeting. Tulane University. October 2025.

(♯) Minicourse on *Metric geometry and shape analysis*. MATRIX workshop “Applications of Topological Data Analysis to Mathematical Biology". MATRIX, University of Melbourne, Australia. November 2024.

The Gromov-Hausdorff distance between spheres. KAIST Math. department. Colloquium. October 2024.

The shape space determined by the Gromov-Wasserstein distance. Sungkyunkwan University. October 2024.

Spatiotemporal persistent homology for dynamic metric spaces. Centre de Recherches Mathématiques (CRM), Montreal. October 2024.

The exact determination of Gromov-type distances between spheres. Centre International de Rencontres Mathématiques (CIRM), Luminy. May 2024.

The exact determination of Gromov-type distances between spheres. Ecole Polytechnique. May 2024.

The Gromov-Hausdorff distance between spheres. MPI-MiS, Leipzig. December 2023.

The Gromov-Hausdorff distance between spheres. UNESP-Sao Carlos, Brazil. November 2023.

The Gromov-Hausdorff distance between spheres. University of Pittsburgh. Colloquium. October 2023.

Curvature sets and curvature measures over persistence diagrams. ICIAM. Tokyo. August 2023.

(∗∗) *The Gromov-Hausdorff distance between spheres*. TDA Week. Kyoto. August 2023.

(∗∗) *The Gromov-Hausdorff distance between spheres*. FoCM Workshop on Computational Geometry and Topology. June 2023.

Curvature Sets Over Persistence Diagrams. Institute for Mathematical and Statistical Innovation (IMSI) workshop on “Randomness in Topology and its Applications”. March 2023.

The Gromov-Hausdorff distance between spheres. University of Wisconsin–Madison (UW-Madison) Colloquium. February 2023.

La persistencia homológica de espacios métricos y su estabilidad. Universidad de la República, Uruguay. December 2022.

Gromov-type distances between spheres. Oberwolfach workshop on “Heat Kernels, Stochastic Processes and Functional Inequalities”. November 2022.

Gromov-type distances between spheres. University of Bonn workshop on “From Dirichlet Forms to Wasserstein Geometry”. October 2022.

(∗∗) *The Gromov-Hausdorff distance between spheres*. ATMCS 2022. Oxford. June 2022.

The ultrametric Gromov-Hausdorff distance. Banff workshop on “Geometry & Learning from Data”. October 2021.

The Gromov-Wasserstein distance and distributional invariants of datasets. MPI-MiS seminar, Leipzig. September 2021.

Some Geometry Rigidity Results via Distance Distribution. SIAM Algebraic Geometry conference. August 2021.

The ultrametric Gromov-Hausdorff distance. AMS Sectional Meeting AMS Special Session "Graphs in Data Science". March 2021.

The ultrametric Gromov-Hausdorff distance. University of Minnesota Data Science seminar. February 2021.

Vietoris-Rips Persistent Homology, Injective Metric Spaces, and The Filling Radius. Fields Institute workshop on Topological Data Analysis. June 2020.

(∗∗) *La distancia de Gromov-Wasserstein e invariantes distribucionales de conjuntos de datos*. Coloquio Uruguayo de Matemáticas. Universidad de la República. Uruguay. Montevideo. December 2019.

(∗) *Inverse problems in metric measure geometry*. NeurIPS workshop on Optimal Transport. Vancouver. December 2019.

The Gromov-Hausdorff distance between ultrametric spaces. FSU Topology seminar. November 2019.

The Gromov-Wasserstein distance and distributional invariants of datasets. SFB Colloquium. Technical University of Munich (TUM). November 2019.

(♯) *The shape space induced by the Gromov-Wasserstein distance*. Lecture series at Universität Göttingen. October 2019.

The Gromov-Wasserstein distance and distributional invariants of datasets. MaD Seminar. New York University (NYU), September 2019.

Metric graph approximation of geodesic metric spaces. Summer Topology Conference. Johannesburg, South Africa. July 2019.

Metric graph approximation of geodesic metric spaces. UNESP, Brazil. June 2019

Stable invariants for Dynamic Metric Spaces. Conference on Geometric Data Analysis. University of Chicago. May 2019.

The Gromov-Wasserstein distance and distributional invariants of datasets. Oberwolfach workshop on " Statistical and Computational Aspects of Learning". May 2019.

Time Dependent Data, Persistence, and Stability. Workshop on "Topological Data Analysis, with Applications". University of Western Ontario. May 2019.

Metrics on the collection of dynamic shapes. Workshop on Shape Analysis, Stochastic Mechanics and Optimal Transport. Banff International Research Station (BIRS), Banff. December 2018.

Stable Signatures for Dynamic Graphs and Dynamic Metric Spaces via Zigzag Persistence. Brazilian Topology Meeting. Niteroi. August 2018.

Stable Signatures for Dynamic Graphs and Dynamic Metric Spaces via Zigzag Persistence. Abel Symposium. Geiranger. June 2018.

(*) *Algebraic methods for characterizing dynamic networks*. OSU–Universidade de São Paulo (USP) joint mathematics conference. Sao Paulo. March 2018.

Stable signatures for dynamic metric spaces via persistent homology. Oberwolfach workshop on Statistics for Data with Geometric Structure. January 2018.

Analysis of dynamic networks via persistent homology. ICERM workshop on "Geometry and Topology of Data". December 2017.

The (persistent) topology of finite metric spaces. Florida State University. Math Department Colloquium. September 2017.

Persistent homology of asymmetric networks. Encontro Regional de Topologia 2017. Sao Paulo State, Brazil. October 2017.

Persistent homology of asymmetric networks. Banff workshop on "Topological Data Analysis: Developing Abstract Foundations". August 2017.

Persistent Homology of Asymmetric Networks. Foundations of Computational Mathematics Conference. Workshop on "Computational Geometry and Topology". Barcelona, July 2017.

Persistent homology of asymmetric networks. Workshop on Geometry and Shape Analysis in Biological Sciences. Institute for Mathematical Sciences. National University of Singapore. June 2017.

Persistent Homology of Asymmetric Networks. Center for Nonlinear Analysis (CNA)/Ki-Net Workshop: Dynamics and Geometry from High Dimensional Data. Carnegie Mellon University (CMU). March 2017.

Multiscale Covariance Fields and Shape Characterization in Euclidean Spaces. Workshop on "Mathematics of Shapes and Applications". Institute for the Mathematical Sciences. National University of Singapore. July 2016.

Multiscale Mapper: Topological Summarization Via Codomain Covers. SIAM Imaging Science workshop on "Topology and Geometry Across Scales". May 2016.

Multiscale Covariance Fields and Shape Characterization in Euclidean Spaces. SIAM Imaging Science workshop on "Theoretical and Computational Aspects of Geometric Shape Analysis ". May 2016.

The shape space defined by the Gromov-Wasserstein distance. Pattern Theory Seminar. Brown University. March 2016.

Hierarchical clustering on asymmetric networks. Institute for Advanced Study (IAS)–Rutgers–University of Pennsylvania (UPenn) Workshop on Topology: Identifying Order in Complex Systems, IAS, November 2015.

Persistent Homology of Finite Metric Spaces and its Stability. Encontro Regional de Topologia de Sao Paulo. Universidade Federal de São Carlos (UFSCar) and UNESP, Sao Carlos, SP, October 2015.

The shape space defined by the Gromov-Wasserstein distance. Seminario de Geometria, Mathematics Department, Universidade de Sao Paulo. October, 2015.

A spectral notion of distance between shapes. Center for Nonlinear Analysis Seminar, CMU. April 30, 2015.

The shape space defined by the Gromov-Wasserstein distance. Geometry seminar, Mathematics Department, Pennsylvania State University (Penn State). April 29, 2015.

The classification of clustering schemes. The geometry luncheon seminar, Mathematics Department, Penn State. April 29, 2015.

Hierarchical clustering on asymmetric networks. Institute for Advanced Study. The Hong Kong University for Science and Technology. Workshop "Functoriality in Data analysis". April 13–17, 2015.

Persistence Diagrams of Metric Measure Spaces. Hausdorff Institute for Mathematics. University of Bonn. New Trends in Optimal Transport. March 2–6, 2015.

The shape space defined by the Gromov-Wasserstein distance. Erwin Schroedinger Institute. Infinite-dimensional Riemannian geometry with applications to image matching and shape analysis: week 4, The Geometry of Optimal Transport. February 2–6, 2015.

The shape space defined by the Gromov-Wasserstein distance. European Research Council workshop on Optimal Transportation. Scuola Normale Superiore di Pisa. October 27–31, 2014.

Metric geometry and Data Analysis. Southeast Geometry Symposium. Knoxville, Tennessee. October 23, 2014.

The shape space defined by the Gromov-Wasserstein distance. University of Bologna. July 16, 2014.

A spectral notion of distance between shapes. MBI. ICIAM Scientific Workshop. May 15–16, 2014.

The shape space defined by the Gromov-Wasserstein distance. University of Georgia Geometry Seminar. March 21, 2014.

Classifying Clustering Schemes. IMA workshop "Topological Systems: Communication, Sensing, and Actuation". March 5, 2014.

Curvature sets over persistence diagrams. Statistical and Applied Mathematical Sciences Institute (SAMSI). LDHD program (low dimensional structure in high dimensional data). February 3–7, 2014.

Curvature sets over persistence diagrams. IMA-UMN. Topological Data analysis. October, 2013.

The Gromov-Wasserstein distance and lower bounds. Luminy, France. CIRM, October, 2013.

Curvature sets over persistence diagrams. University of Bremen, Germany. Applied and Computational Topology. July 2013.

Curvature sets over persistence diagrams. Bedlewo, Poland. Applied Topology. July 2013.

The shape space defined by the Gromov-Wasserstein distance. Workshop on Geometry and Statistics in bioimaging: manifolds and stratified spaces. October 8–12, 2012 at Sandbjerg Estate, Sonderborg, Denmark.

Persistence homology and metric geometry. Banff workshop on Topological Data Analysis and Machine Learning Theory, October 14–19, 2012.

Metric Geometry and Persistent Homology. ATMCS 2012 (Algebraic topology methods in computer science). Edinburgh, July 2012.

Lower Bounds for the Gromov-Hausdorff distance Using Persistent Topology. SIAM Conference on Applied Algebraic Geometry. Raleigh, North Carolina. October 6–9, 2011.

The Classification of Clustering Schemes. National ICT Australia (NICTA), SML seminar series. September 29, 2011. NICTA, Canberra.

(∗∗) *Metric Geometry in Data Analysis*. CAIP 2011: 14th International Conference of Computer Analysis of Images and Patterns. Sevilla, Spain, August 29–31, 2011.

The classification of clustering schemes. Dagstuhl seminar on Mathematical and Computational Foundations of Learning Theory. Schloss Dagstuhl, Germany, July 17–22, 2011.

Some properties of Gromov-Hausdorff distances. Shape Focused Research group meeting. Organized by Darryl Holm. Imperial College. London, UK. May 2011.

The Gromov-Wasserstein distance and object matching. Center for Nonlinear Science seminar series, CMU Mathematics department, October 12, 2010.

Metric geometry in shape analysis. AMS 2010 Fall Western Section Meeting, University of California, Los Angeles (UCLA), October 9, 2010.

Metric geometry in shape matching. NCSU, Electrical Engineering Department. Raleigh, North Carolina, September 16, 2010.

Some metric geometry ideas for object matching. ACCV 2010, area chair Colloquium. National Institute of Informatics, Tokyo, Japan, September 1, 2010.

(∗) *Metric Geometry for object matching*. NORDIA-CVPR 2010 workshop. San Francisco, USA. June 18, 2010.

(∗∗) *Stability of Persistence Homology signatures arising from Vietoris-Rips simplicial constructions*. Data Topology (Topología de Datos). Santiago de Compostela, Spain. May 21–22, 2010.

Spectral Gromov-Wasserstein Distances for Shape Matching and the Role of Scale. SIAM IS 2010 workshop on "Local Scales: Theory and Applications". Chicago, April 14, 2010.

Characterization, stability and convergence of hierarchical clustering schemes. University of Washington, Statistics Department. Seattle, Washington, February 8, 2010.

Gromov-Wasserstein stable signatures for object matching and the role of persistence. Workshop on Recent Advances on Topological and Geometric Data Analysis. Institut des Systemes complexes. Paris, France, July 8–10, 2009.

A spectral notion of Gromov-Wasserstein distances. Applied analysis seminar, Yale University, October 12, 2009.

Gromov-Wasserstein distances for object matching. École normale supérieure de Cachan (ENS-Cachan), Centre de mathématiques et de leurs applications (CMLA). Paris, France, June 18, 2009.

Gromov-Wasserstein distances for shape matching and connections with other approaches. Centre de recherche en mathématiques de la décision (CEREMADE), Paris-Dauphine, June 16, 2009.

Shape Matching using L^p-Gromov-Hausdorff distances. Computer Science department seminar, Johns Hopkins University. April 27, 2009.

Some metric geometry ideas for shape matching. Annapolis Shape Meeting. Organized by Laurent Younes, Peter Michor and David Mumford. Annapolis, MD. April 20–22, 2009.

A metric geometry approach to object matching. Workshop on Data Analysis using Computational Topology and Geometric Statistics. BIRS station, Banff, Canada. March 8–13, 2009.

L^p-Gromov-Hausdorff distances for Shape Comparison. Applied Math. Colloquium, UCLA, October 29, 2008.

Stability of Clustering Methods. Workshop on Computational Algebraic Topology. Mathematisches Forschungsinstitut Oberwolfach. Oberwolfach, Germany, July 2008.

L^p-Gromov-Hausdorff distances for Shape Comparison. Workshop on Geometry and Statistics of Shapes, Hausdorff Institute of Mathematics, University of Bonn, Germany, June 2008.

L^p-Gromov-Hausdorff distances for Shape Comparison. Geometric Modeling and Industrial Geometry Group seminar, Vienna University of Technology (TU Wien), Vienna, Austria, June 17, 2008.

Isometry invariant shape characterization. Workshop on 3D Shape Characterization, University of Konstanz, Germany December 18, 2007.

Two tools for multidimensional data analysis: Mapper and metric training. DARPA Topological Data Analysis meeting. San Diego, December 11, 2007.

Some ideas for the Comparison of Shapes given as Point Cloud Data. Applied Math. Group seminar at Stanford University, February 16, 2007.

(∗) *Qualitative Analysis of Evoked data from V1*. International Symposium “Vision by Brain and Machines". Montevideo, Uruguay, November 13–17, 2006.

Analysis of Visual Cortex Data. DARPA Topological Data Analysis meeting, Santa Barbara, May 9, 2006.

Some idea for comparing shapes under invariances. DDMA Speaker Series. Los Alamos National Laboratory, New Mexico, April 13, 2006.

Comparing Point Clouds: Theory and Algorithms. SIAM Conference on Imaging Science, Salt Lake City, Utah, May 3–7, 2004.

PDEs on and Between Manifolds for Image Science Applications. SIAM Conference on Imaging Science, Salt Lake City, Utah, May 3–7, 2004.

Distance Functions on submanifolds of $\mathbb{R}^d$ and Point Clouds. SIAM 50th Anniversary and Annual Meeting, Philadelphia, Pennsylvania, July 8–12, 2002.

Conference and Workshop Presentations

Improved Error Bounds for Tree Representations of Metric Spaces. ITA 2017. February 2017.

Hierarchical clustering methods on asymmetric networks. ITA 2016, San Diego, CA, February 2016.

Multiscale Mapper: A framework for topological summarization. SODA 2016, Arlington, VA, January 2016.

Informational robustness of the hippocampal spatial map. SACNAS 2015, Washington DC.

Metric Structures on Networks and Applications. With S. Chowdhury. Allerton, 2015.

Some properties of Gromov-Hausdorff distances. Eighth International Conference on Mathematical Methods for Curves and Surfaces. Oslo, June 28–July 3, 2012.

Stability of Persistent Homology Invariants. 55th Australian Mathematical Society Meeting. Wollongong, September 26–29, 2011.

A study of spectral methods for surface and data matching from the Gromov-Wasserstein point of view. Seventh International Conference on Curves and Surfaces. Avignon, France. June 24–30, 2010.

Spectral Gromov-Wasserstein distances for shape matching. Workshop on Non-Rigid Shape Analysis and Deformable Image Alignment (NORDIA) ICCV’09. Kyoto, Japan. September 27, 2009.

Informational Robustness of the Hippocampal Spatial Map: Topological Analysis. Workshop on Methods of Information Theory in Computational Neuroscience, CNS 2009 (Computational Neuroscience). Berlin, Germany, July 22 and 23, 2009.

Gromov-Hausdorff stable signatures for shapes using persistence. Symposium on Geometry Processing 2009 (SGP). Berlin, Germany, July 15–17, 2009.

Topological stability of the hippocampal spatial map. COSYNE 2009. Salt Lake city, Utah, February 26–March 1, 2009.

Gromov-Hausdorff distances in Euclidean Spaces. Workshop on Non-Rigid Shape Analysis and Deformable Image Alignment (NORDIA) CVPR'08. Alaska, June 27, 2008.

Topological Structure of Population Activity in Primary Visual Cortex. With G. Carlsson, D. Ringach, G. Sapiro and G. Singh. COSYNE 2008. Salt Lake City, Utah, February 28–March 2, 2008.

On the use of Gromov-Hausdorff distances for Shape Comparison. PBG 2007 (Symposium on Point Based Graphics). Prague, Czech Republic, September 2–3, 2007.

Topological Methods for the Analysis of High Dimensional Data Sets and 3D Object Recognition. PBG 2007 (Symposium on Point Based Graphics). Prague, Czech Republic, September 2–3, 2007.

Comparing Point Clouds. SGP 2004 (Symposium in Geometric Processing). Nice, France, July 8–10, 2004.

Solving Variational Problems and PDEs mapping into General Target Manifolds. In 2nd IEEE Workshop on Variational, Geometric and Level Set Methods in Computer Vision In Conjunction with the 9th IEEE International Conference in Computer Vision. Nice, France, October 12, 2003.